ModelIt!

Teacher's Guide

# Engineering Earthquake-Proof Skyscrapers

Modeling the Physics of Structural Survival Under Seismic Forces

## Quick Reference

| | |
|---|---|
| **Grade Level** | Grade 10L3 |
| **NGSS Standard** | HS-ESS2-2, HS-PS2-1 |
| **Time** | 45–60 minutes (core lesson) |
| **Materials** | Student Activity Pack, devices with ModelIt! access |
| **Prep** | Print activity packs, test ModelIt! on student devices |
| **Grouping** | Partners or small groups (2–4) |

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# 1 NGSS Standards & CAST Alignment

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**Performance Expectation: HS-ESS2-2, HS-PS2-1**

Analyze geoscience data to make the claim that one change to Earth's surface can create feedbacks that cause changes to other Earth systems. Analyze data to support the claim that Newton's second law of motion describes the mathematical relationship among the net force on a macroscopic object, its mass, and its acceleration.

## Three-Dimensional Learning

| | |
|---|---|
| Science Practice | **Analyzing and Interpreting Data** — Students analyze seismic wave data and structural response data to identify relationships between ground motion characteristics, building properties, and damage outcomes. |
| Disciplinary Core Idea | **ESS2.B Plate Tectonics and Large-Scale System Interactions / PS2.A Forces and Motion** — Students apply Newton's second law (F=ma) to understand how seismic accelerations generate forces proportional to building mass, and how plate tectonics generates the seismic waves. |
| Crosscutting Concept | **Structure and Function** — Students analyze how the structural design of a building (its shape, materials, connections, dampers) determines its function (survival or collapse) during seismic loading. |

## CAST Testing Alignment

**MULTIPLE CHOICE:**

Two buildings of equal height experience the same earthquake. Building A has a natural period of 3.0 seconds and the earthquake's dominant period is 3.1 seconds. Building B has a natural period of 1.2 seconds. According to your model, which building is in greater danger and why?

**CONSTRUCTED RESPONSE:**

Using your model, explain why making a building stronger (higher Material Strength) is not sufficient to make it earthquake-resistant. Address the roles of resonance, flexibility, and energy absorption in your explanation.

# 2 Scoring Rubric

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### Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

| Skill | Expert (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|
| **Identifying Components** | EXPERT<br>Correctly identifies and sorts ALL components as internal or external with detailed reasoning | PROFICIENT<br>Correctly identifies and sorts MOST components with some reasoning | DEVELOPING<br>Identifies components but has difficulty sorting or explaining | BEGINNING<br>Needs significant help identifying components |
| **Establishing Relationships** | EXPERT<br>Accurately describes ALL positive and negative relationships with clear evidence | PROFICIENT<br>Accurately describes MOST relationships with some evidence | DEVELOPING<br>Identifies some relationships but struggles with direction | BEGINNING<br>Needs help understanding how components connect |
| **Running Simulations** | EXPERT<br>Makes accurate predictions AND explains surprising results using evidence | PROFICIENT<br>Makes reasonable predictions and records clear observations | DEVELOPING<br>Runs simulations but predictions or observations are incomplete | BEGINNING<br>Needs help running simulations or recording data |
| **Explaining Results** | EXPERT<br>Uses multiple pieces of evidence to explain HOW and WHY systems interact | PROFICIENT<br>Uses evidence to explain system interactions | DEVELOPING<br>Gives partial explanations with limited evidence | BEGINNING<br>Needs help connecting evidence to explanations |

### Scoring Guidance

**Expert (4):** Student demonstrates deep understanding and can transfer knowledge to new contexts.
**Proficient (3):** Student meets grade-level expectations and shows solid understanding.
**Developing (2):** Student is making progress but needs additional scaffolding or practice.
**Beginning (1):** Student needs significant support; consider using Differentiation strategies (Section 7).

# 3 Background Content Knowledge

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**For Your Confidence**

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Earthquakes generate forces that can accelerate the ground at the same rate as gravity — imagine the floor suddenly tilting 90 degrees. Buildings, which are designed to resist vertical gravity loads, must also survive violent horizontal shaking. The engineering of earthquake-resistant structures is a battle against F=ma: the more mass a building has, the greater the forces it must withstand during acceleration.

## Seismic Wave Physics

Earthquakes generate three types of waves: P-waves (compression, fastest, arrive first), S-waves (shear, side-to-side, more damaging), and surface waves (rolling motion, slowest, most destructive). Each wave type has characteristic frequencies. Most building damage comes from surface waves and S-waves with periods of 0.5-5 seconds — which unfortunately overlaps with the natural periods of most buildings from 3 to 60 stories.

## The Resonance Problem

Every structure has a natural frequency at which it wants to oscillate — like a tuning fork. For tall buildings, this period is roughly 0.1 seconds per story (a 30-story building has a natural period of about 3 seconds). When earthquake waves arrive at frequencies matching the building's natural period, the structure enters resonance: each cycle of ground motion adds energy to the building's oscillation, amplifying forces by 5-10 times. This is why medium-rise buildings (5-15 stories) are particularly vulnerable — their natural periods match the dominant frequencies of most earthquakes.

## Modern Seismic Engineering Strategies

Three primary strategies protect buildings: (1) Base isolation uses flexible bearings between the foundation and structure, allowing the ground to move while the building stays relatively stationary — reducing forces by 50-80%. (2) Structural flexibility through moment-resisting frames and shear walls allows the building to flex without breaking, with a natural period designed to avoid resonance with expected earthquake frequencies. (3) Energy dissipation through tuned mass dampers, viscous fluid dampers, and friction connections absorbs kinetic energy and converts it to heat, reducing peak accelerations on upper floors by 30-50%.

## Design Philosophy: Life Safety vs. Collapse Prevention

Earthquake engineering does not aim for zero damage — that would be prohibitively expensive. Instead, buildings are designed to three performance levels: Immediate Occupancy (minimal damage, building fully usable — for hospitals and fire stations), Life Safety (significant structural damage but no collapse — occupants can evacuate safely — the standard for most buildings), and Collapse Prevention (severe damage, building may be demolished afterward, but it does not kill occupants — the minimum standard). This philosophy accepts that buildings are sacrificial structures that protect human life through controlled damage.

# 4 The LEVER Framework

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### About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

**L** **Locate the System**

Students identify nine key components of the seismic building system: seismic wave intensity, building mass, damper effectiveness, foundation flexibility, resonance frequency, material strength, floor acceleration, evacuation time, and damage index.

**E** **Establish Relationships**

Students map relationships showing that seismic wave intensity and resonance frequency interact to determine floor acceleration, which combines with building mass to generate forces that material strength must resist, while dampers and base isolation reduce the energy reaching the structure.

**V** **Visualize & Model**

Students build a computational model showing the nine-component system with resonance amplification as the critical interaction.

**E** **Evaluate Outcomes**

Students run moderate earthquake, resonance catastrophe, and maximum credible earthquake scenarios to test their model and identify the conditions under which each seismic protection strategy becomes critical.

**R** **Revise & Extend**

Students add soil amplification factor, aftershock vulnerability, or non-structural damage to model more realistic seismic response scenarios.

### Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

| | |
|---|---|
| **At the start of a unit** | Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it. |
| **After a reading or lab** | Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes. |
| **As a “what-if” tool** | When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back. |
| **For peer discussion** | Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence. |
| **As a presentation tool** | Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time. |
| **At the end of a unit** | Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved. |

# 5 Slide-by-Slide Facilitation

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## Slide 1: Cover

2 min

**Say:** *"In 1985, a magnitude 8.0 earthquake hit Mexico City. Buildings of 8-15 stories collapsed everywhere while shorter and taller buildings survived. Why those specific buildings? The answer changed earthquake engineering forever."*

**Watch For**

Show before/after images of the 1985 Mexico City earthquake. The pattern of destruction by building height hooks students into the resonance concept.

## Slide 2: Learning Objectives

3 min

**Say:** *"Today you are engineering buildings that survive the most destructive force on Earth. And the secret is not making them stronger — it is making them smarter."*

**Watch For**

Have students read objectives. Pre-teach 'resonance' using a playground swing analogy: pushing at the natural frequency makes the swing go higher and higher.

## Slide 3: Big Question

3 min

**Say:** *"A skyscraper weighs hundreds of thousands of tons. During an earthquake, F equals ma — all that mass times the ground acceleration equals enormous force. So how does it not fall down?"*

**Watch For**

Quick-write: Students list strategies they think engineers use. Share out. Most will guess 'make it stronger' — set up the surprise that flexibility beats strength.

## Slide 4: LEVER Framework

2 min

**Say:** *"We are building a model of a building's fight against an earthquake — nine variables that determine whether 1,000 people on the upper floors live or die."*

**Watch For**

Preview LEVER steps. Emphasize that this model reveals a counterintuitive truth about what makes buildings survive.

## Slide 5: Activity 1: Components

8 min

**Say:** *"Nine variables. Two you set as conditions — earthquake intensity and foundation type. Seven that the physics determines. Which combinations lead to survival?"*

**Watch For**

Guide students through all nine components. Spend extra time on resonance frequency — this is the key concept that unlocks the model.

## Slide 6: Activity 2: Connections

8 min

**Say:** *"Here is the deadly math: Force equals mass times acceleration. Resonance multiplies acceleration by 5 to 10 times. That means forces that should be survivable become catastrophic."*

**Watch For**

Help students map the resonance amplification pathway. Use the swing analogy: small pushes at the right frequency create enormous motion.

## Slide 7: Activity 3: Simulation

10 min

**Say:** *"Let us find out what kills a building. Spoiler: it is not the biggest earthquake — it is the earthquake at exactly the wrong frequency."*

**Watch For**

Students predict Damage Index for each scenario. The resonance catastrophe scenario should surprise them — moderate earthquake, massive damage. Discuss why.

## Slide 8: Discoveries

5 min

**Say:** *"Now you know the secret: earthquake engineering is not about being the strongest — it is about being the smartest. Avoid resonance, absorb energy, decouple from the ground."*

**Watch For**

Connect model findings to the Mexico City 1985 case: medium-rise buildings collapsed because their natural period matched the earthquake's dominant frequency. Taller and shorter buildings survived.

## Slide 9: STEM Challenge

12 min

**Say:** *"Your city just approved a 60-story tower 8 km from an active fault. Design the seismic system that keeps everyone alive during a magnitude 7.5 earthquake."*

**Watch For**

Groups design comprehensive seismic protection systems. Test concepts against model scenarios. Present with engineering justification from simulation data.

# 6 Answer Key & Expected Responses

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## Component Sorting

- **EXTERNAL:** Seismic Wave Intensity, Foundation Flexibility
- **INTERNAL:** Building Mass, Damper Effectiveness, Resonance Frequency, Material Strength, Floor Acceleration, Evacuation Time, Damage Index

## Relationships

- **Seismic Wave Intensity to Floor Acceleration:** POSITIVE (+) — Higher seismic wave intensity at the foundation drives greater accelerations throughout the building structure. Floor accelerations are typically amplified toward the top of the building, with upper floors experiencing 2-3 times the ground-level acceleration.
- **Resonance Frequency to Floor Acceleration:** AMPLIFYING — When the dominant frequency of seismic waves approaches the building's natural frequency, floor accelerations are amplified by a factor of 5-10 through resonance. This amplification is the single most destructive phenomenon in earthquake engineering.
- **Floor Acceleration to Damage Index:** POSITIVE (+) — Higher floor accelerations generate greater forces on structural members (F=ma), increasing the stress on connections, columns, and beams. When these forces exceed material strength, members fail, connections fracture, and the Damage Index rises toward collapse.

## Key Discoveries

- Resonance is the primary killer in earthquakes — buildings collapse not because the ground motion is too strong but because the building amplifies the motion to destructive levels when frequencies match
- Modern seismic engineering uses a three-layer defense: base isolation decouples from ground motion, flexible design avoids resonance, and dampers absorb energy that gets through
- Building mass is a double-edged sword: heavier buildings have lower natural frequencies (avoiding resonance with typical seismic waves) but experience greater inertial forces at any given acceleration
- No building can be made truly earthquake-proof — engineers design for graceful failure, where the structure absorbs damage in planned locations while protecting life safety and evacuation routes

# 7 Collaborative Learning Protocols

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### Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

### Model Gallery Walk

**When:** After students build models (LEVER Phase V)
**Setup:** Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
**Protocol:** Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
**Why it works:** Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

### Scientist Fishbowl

**When:** During simulation discussion (LEVER Phase E)
**Setup:** 4-5 students sit in center circle. Rest of class observes with clipboards.
**Protocol:** Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
**Why it works:** Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

### Component Jigsaw

**When:** During component analysis (LEVER Phase L)
**Setup:** Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
**Protocol:** Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
**Why it works:** Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

### Prediction Debate

**When:** Before running simulations (LEVER Phase E)
**Setup:** Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
**Protocol:** Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
**Why it works:** Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

### Pair Programming for Models (Digital Collaboration)

**When:** Throughout model building
**Setup:** Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
**Protocol:** Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
**Why it works:** Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

### Quick Protocol Selection Guide

| | |
|---|---|
| **Short on time (5 min)** | Prediction Debate — gets bodies moving, generates energy fast |
| **Full period available** | Component Jigsaw — deep collaboration, every student contributes |

| | |
|---|---|
| **After simulation day** | Model Gallery Walk — peer feedback, multiple perspectives |
| **Advanced discussion** | Scientist Fishbowl — builds academic discourse skills |
| **Every day default** | Pair Programming — transforms individual work into dialogue |

# 8 Differentiation & Extensions

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### Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

### Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

### English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

### Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

# 9 Common Misconceptions

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### Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

| STUDENTS MAY THINK | SCIENTIFIC UNDERSTANDING |
|---|---|
| "Stronger buildings are safer in earthquakes" | This is one of the most dangerous misconceptions in earthquake engineering. A perfectly rigid, extremely strong building transmits 100% of ground acceleration to every floor. If that acceleration hits the building's resonance frequency, forces amplify to destructive levels regardless of material strength. Flexible buildings that deform elastically absorb and dissipate seismic energy, reducing peak forces. Base-isolated buildings decouple entirely. The lesson: flexibility and energy absorption protect buildings more than raw strength. |
| **STUDENTS MAY THINK** | **SCIENTIFIC UNDERSTANDING** |
| "Earthquakes destroy buildings by shaking them apart" | Earthquakes primarily destroy buildings through resonance amplification and connection failure, not through raw shaking intensity. Most structural collapses occur because the building's natural frequency matched the earthquake's dominant frequency, amplifying motion to 5-10 times the ground movement. Failure typically begins at connections — where beams meet columns — not in the members themselves. A well-connected, flexible building can survive extreme shaking; a poorly connected, rigid building can collapse in a moderate earthquake. |
| **STUDENTS MAY THINK** | **SCIENTIFIC UNDERSTANDING** |
| "If a building survives one earthquake, it will survive the next one" | Every earthquake causes some degree of damage, even if invisible. Micro-cracks in concrete, loosened bolts, fatigued steel connections, and permanent deformation all reduce the building's remaining capacity. A building that barely survived a moderate earthquake may collapse in an aftershock that is weaker than the initial quake. This is why post-earthquake inspections are critical — a building may look fine from outside while its structural reserves have been depleted. The model shows how Damage Index does not reset between seismic events. |